

Western University

Faculty of Engineering

Mechatronic Systems Engineering Program

MSE 2202 – Introduction to Mechatronic Design

Product Design Report

Scavenger Robot Project: Welcome to Servo Buster

Lab 003 – Team 1

Group Members

Youssef El Kabat — 251370760

Marc Masgoret — 251412418

Eric Halim — 251241625

Dalia Salamah — 251402061

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Executive Summary

This report presents the design, development, and validation of an autonomous scavenger system for the MSE 2202 term project. The objective was to design a compact, self-powered system capable of autonomously collecting erasers from an unmarked arena, returning them to a base station, and sorting target green erasers from non-target objects within a 120-second time limit.

The final design consisted of two coordinated subsystems: a mobile collection robot and a stationary sorting subsystem at the base station. The robot used rear-wheel drive, a scoop-and-brush intake, IMU-assisted navigation, and ultrasonic-based return positioning. The stationary sorter used a dual vibratory feeder, a TCS34725 colour sensor, and a servo-actuated chute to classify deposited erasers under controlled conditions. This architecture was selected because it reduced the mechanical complexity of the robot while improving sorting reliability.

The design evolved significantly through testing and refinement. Major improvements included reducing chassis weight, redesigning the scoop to lower servo loading, improving alignment at the base station, and replacing the initial rotary sorting concept with a more reliable vibratory feeder. Validation testing showed that the final system completed the full sequence of collection, return, unloading, and sorting in an average of 85.7 s across the final 12 runs. Sorting accuracy reached 90–92% under controlled trials.

The main conclusion is that the final design provided a feasible and technically sound solution to the project requirements. Future development should focus on improving object retention, return-path accuracy, and single-file object feeding at the sorting station to further increase reliability and sorting accuracy.

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1-Introduction

1.1 Design Problem

Autonomous robotic systems are increasingly deployed in environments where human operation is unsafe, inefficient, or impractical — including hazardous material recovery, remote inspection, industrial automation, and search-and-retrieval applications. In these contexts, the system must do more than traverse an environment; it must reliably identify, collect, transport, and process objects of interest under real operating conditions. The design challenge addressed in this project was the development of a compact, self-powered autonomous scavenger system capable of completing a full collection-and-sorting task without human intervention.

Per the project requirements, the system had to depart from a defined base location, gather valuable objects from the ground within an operating area of up to 15 m², return to the base station, and deposit only the target objects into a collection container — all within a maximum of 120 seconds. The system was required to operate fully autonomously, with no painted lines, installed guides, tethered power supplies, or external controllers, and had to be implemented using an ESP32-based microcontroller. The complete assembly also had to remain compact enough to fit within an MSE locker without disassembly. The operating surface was assumed to be relatively flat and bounded primarily at its perimeter, with no major internal obstacles.

Beyond these stated requirements, the actual engineering problem was considerably broader. A successful design had to simultaneously balance mobility, collection efficiency, object retention, sorting reliability, manufacturability, mechanical simplicity, power constraints, and robustness under real operating conditions. These competing demands motivated the adoption of a two-subsystem architecture: a mobile scavenger robot responsible for navigation and object collection, paired with a stationary sorting subsystem located at the base station. Although physically separate, the two subsystems were designed to function as one unified solution. This architecture reduced the mechanical complexity and sensing burden

on the mobile platform while improving the feasibility, calibration accessibility, and overall reliability of the sorting stage.

1.2 Background Information

This project required the integration of several core mechatronic disciplines, including autonomous navigation, ground-level object collection, and colour-based object classification. A review of relevant approaches informed the final design decisions in each of these areas.

For small autonomous robots operating in bounded indoor environments, common navigation strategies include dead reckoning, inertial heading control, line-following, infrared beacon homing, and vision-based localization. Dead reckoning using wheel encoders is straightforward to implement but accumulates positional error over time due to wheel slip and mechanical tolerances. Line-following eliminates drift but requires pre-installed guides, which conflicts directly with the project constraints. Infrared beacon homing offers a practical middle ground: it provides directional feedback toward a known target without requiring external infrastructure, making it well-suited to the return-to-base phase of this task. For area coverage, open-loop timed sweep patterns — while less precise than closed-loop localization — are robust, computationally simple, and sufficient for bounded flat environments where obstacle density is low.

Ground-level object collection mechanisms for small robots include claws, plows, rotating brushes, conveyor belts, scoops, and vacuum-based systems. Claw mechanisms allow targeted manipulation but introduce actuation complexity and reduce collection throughput. Vacuum and conveyor systems support continuous pickup but demand additional power, volume, and fabrication effort. Brush-and-scoop arrangements represent an effective compromise: they are mechanically simple, tolerant of variability in object position, and capable of gathering multiple objects rapidly — properties that are well-aligned with the time-constrained scavenging task defined in this project.

Sorting of small, coloured objects can be achieved using photodiode-based colour sensing, RGB sensors, or machine vision. RGB colour sensors such as the TCS34725 provide reliable hue discrimination at low cost and with minimal computational overhead, making them appropriate for embedded systems with limited processing resources. Sorting can be implemented either onboard the mobile robot or via a stationary system at the base station. Onboard sorting minimises transfer steps but increases the robot's weight, sensing complexity, and packaging constraints. A stationary sorter, by contrast, allows objects to be presented to the sensor in a controlled, repeatable manner, improving classification accuracy and simplifying calibration. For an academic prototype operating under a fixed budget and timeline, the stationary approach offers a more reliable and maintainable solution.

More sophisticated approaches — including encoder-based odometry with sensor fusion, simultaneous localization and mapping (SLAM), or convolutional neural network-based vision classification — could improve adaptability and spatial precision. However, these methods introduce significant hardware cost, computational overhead, and development complexity that were inconsistent with the available resources and project timeline. The final design therefore prioritised practical robustness and subsystem modularity over algorithmic sophistication.

1.3 Project Objectives and Scope

The objective of this project was to design, fabricate, and validate a complete autonomous scavenger system capable of collecting target objects from an unmarked floor area and ensuring that only the desired objects were deposited into the final collection container. Functionally, the completed system had to:

1. Navigate autonomously through the operating area without external guides or remote control
2. Collect multiple objects from the ground efficiently within the mission time limit
3. Transport collected objects back to the base station
4. Sort the collected objects such that only target items were retained in the final collection stage

Expressed in quantitative engineering terms, the system was required to complete the full task within 120 seconds, operate entirely on the onboard power bank included in the MSE 2202 Lab Kit, use an ESP32-based microcontroller for all control functions, remain within the physical envelope of an MSE locker without disassembly, and be fabricated within a combined additional component budget of \$50. The project scope encompassed concept generation and selection, CAD-based design development, prototype fabrication, electrical and software integration, and performance testing against the defined specifications.

These objectives were achievable within the available timeframe because the final architecture deliberately decomposed the overall mission into two cooperating subsystems — a mobile robot optimised for navigation and collection, and a stationary sorter optimised for classification and final deposition — each of which could be developed, tested, and refined in parallel.

2. Discussion

2.1 Concept Generation and Selection

A structured concept generation process was used to develop the final system architecture. Rather than selecting a complete robot concept immediately, the overall problem was first decomposed into major functional subsystems: drivetrain, navigation and heading control, environment sensing, search strategy, object collection, sorting location, sorting mechanism, homing, object deposition, and structural packaging. Multiple concepts were generated for each subsystem independently to explore a broad design space before committing to an integrated solution. This approach ensured that final concept selection was grounded in engineering trade-offs rather than intuition alone. Alternatives considered during the Milestone 1 concept generation phase included front-wheel, all-wheel, and three-wheel drive layouts; open-loop, encoder-based, and IMU-based motion control; random-walk, lawnmower, and spiral coverage strategies; onboard versus base-station sorting; and multiple collection and deposition mechanisms.

At the subsystem level, each selected concept was chosen on the basis of its balance between performance, reliability, simplicity, and feasibility within the course timeline and budget. For locomotion, a four-wheel all-wheel-drive layout was identified as the preferred concept in Milestone 1 because it offered improved traction, stability, and turning control while pushing a front-mounted collection mechanism. For navigation, IMU-based heading feedback was selected over open-loop PWM control and encoder-based speed matching because it improved directional consistency without introducing the encoder mounting and tuning challenges associated with wheel-speed feedback on a compact platform. A single ultrasonic sensor was selected for obstacle detection on the grounds that the competition arena was assumed to be relatively open and flat, making a simple single-axis sensing configuration sufficient for safe autonomous operation. A lawnmower-style zig-zag coverage strategy was chosen because it offered more predictable area coverage than a random walk and lower implementation complexity than a spiral or perimeter-inward strategy. For sorting, a stationary base-station sorter was preferred over onboard sorting because it reduced the mass and mechanical complexity of the mobile robot while simultaneously creating more controlled and repeatable sensing conditions for object colour classification. Within the base-station sorting category, two distinct mechanisms were evaluated: a motorised rotary indexing disc with radial ridges intended to capture and index erasers one at a time, and a dual vibratory feeder with a servo-actuated bifurcated chute. The rotary disc was rejected after physical testing revealed that the compliant and irregular surface geometry of the erasers prevented consistent single-object indexing, making the vibratory feeder the stronger candidate.

These selected subsystem concepts were then assembled into three complete system-level concepts for comparative evaluation. Concept A used a scoop with active intake, infrared beacon homing, and a gravity-based ramp sorter with a servo-controlled gate. Concept B used a six-wheel skid-steer platform, claw-based pickup, onboard sorting, and encoder-based homing. Concept C used a four-wheel-drive chassis, scoop-based collection with active brush intake, IMU-assisted navigation, and a stationary base sorting station with colour sensing and active servo diversion. Concept C was selected because it provided

the strongest overall balance between collection efficiency, sorting reliability, navigation performance, and integration feasibility. Compared with Concept A, it reduced sensitivity to gravity timing and trapdoor alignment by using a more controlled vibration-assisted transfer and diverter-based sorting process. Compared with Concept B, it avoided the high mechanical and programming complexity associated with claw-based pickup and onboard real-time object discrimination. A Go/No-Go screening eliminated Concepts A and B due to unacceptable risk in sorting reliability, homing robustness, and mechanical complexity, while Concept C passed all mandatory screening criteria. This result was subsequently confirmed by a weighted decision matrix, in which Concept C received the highest total score of 4.20, compared with 3.15 for Concept A and 2.75 for Concept B.

From an engineering standpoint, the selection of Concept C was justified primarily because it separated the overall mission into two cooperating but independently optimised subsystems. The mobile robot was optimised for motion, collection, and return-to-base behaviour, while the stationary sorter was optimised for controlled object presentation and classification. This separation reduced the number of interacting mechanisms that had to function simultaneously on the robot, lowered the risk of onboard jamming or sensing interference, and improved the likelihood of completing the full mission within the 120-second time constraint. The architecture was also more realistic to fabricate, debug, and refine within the available time and course resources than either alternative. The concept selection process therefore identified not only a feasible design direction but also demonstrated why it represented the most practically sound and technically balanced solution among the candidates considered.

While Concept C was selected as the foundation for further development, the final prototype was not a direct one-to-one implementation of the original conceptual design. As prototyping and subsystem testing progressed, several elements of the concept were refined to improve manufacturability, reduce weight, simplify integration, and increase overall reliability. These refinements included changes to the drivetrain layout, chassis structure, scoop geometry, unloading method, and collection sequence. Concept C should

therefore be understood as the selected design direction, while the final prototype represents its optimised and experimentally refined implementation.

Table 1: Go/No-Go screening matrix applied to the three system-level design concepts. Concepts A and B were eliminated due to unacceptable risk in sorting reliability, homing robustness, and mechanical complexity. Concept C passed all mandatory screening criteria and was selected for further development.

#	Screening Criteria	Concept A	Concept B	Concept C
1	Can collect multiple erasers efficiently	GO	NO-GO	GO
2	Meets 120-second time constraint	GO	NO-GO	GO
3	Sorting reliability acceptable	NO-GO	MAYBE	GO
4	Mechanical complexity manageable	MAYBE	NO-GO	GO
5	Programming complexity manageable	GO	NO-GO	GO
6	Jamming risk acceptable	MAYBE	NO-GO	GO
7	Weight within acceptable range	GO	NO-GO	MAYBE
8	Fabrication feasible within timeline	GO	NO-GO	GO
9	Reliable homing achievable	NO-GO	MAYBE	GO
10	Debugging practical within project time	GO	NO-GO	GO
	Final Screening Decision	NO-GO	NO-GO	GO

Table 2: Weighted decision matrix comparing Concepts A, B, and C across seven engineering criteria.

Weights were assigned based on relative importance to mission success. Concept C achieved the highest weighted total score of 4.20, confirming the Go/No-Go screening result.

#	Criterion	Weight	Concept A	W.A	Concept B	W.B	Concept C	W.C
1	Collection Efficiency	0.20	3	0.60	2	0.40	5	1
2	Sorting Accuracy & Reliability	0.20	3	0.60	4	0.80	4	0.80
3	Autonomous Navigation Performance	0.15	3	0.45	4	0.60	5	0.75
4	Mechanical Complexity	0.15	3	0.45	2	0.30	3	0.45
5	Risk of Jamming / Failure	0.15	3	0.45	2	0.30	4	0.60
6	Manufacturability & Integration Feasibility	0.10	4	0.40	2	0.20	4	0.40
7	Stability & Mass Distribution	0.05	4	0.20	3	0.15	4	0.20
	Total	1.00		3.15		2.75		4.2

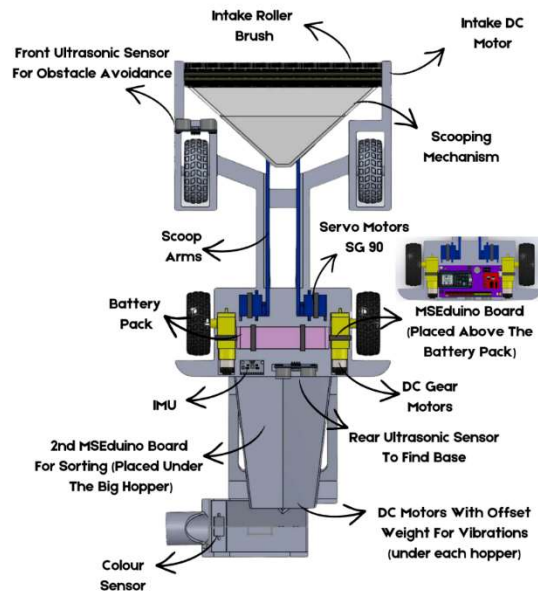


Figure 1: Annotated SolidWorks top-view assembly drawing of the final scavenger robot, identifying all major subsystems including the intake roller brush, scooping mechanism, scoop arms, SG90 servo motors, DC gear motors, IMU, front and rear ultrasonic sensors, battery pack, MSEduino control boards, and vibration motors.

2.2 Detailed Design and System Implementation

The final design comprised two coordinated subsystems: a mobile collection robot and a stationary sorting station, each optimised for its specific function within the overall mission. The detailed design process focused on refining both subsystems into a manufacturable and functional final product that could be built within the available time, budget, and fabrication constraints while operating reliably as one integrated system.

Mobile Robot — Drivetrain and Chassis

The mobile robot was designed around a compact wheeled chassis with a front-mounted collection mechanism and a rear section housing the drive system, electronics, and structural support. In the refined final design, the robot used two powered rear wheels driven by DC gear motors, with passive front

support wheels providing stability. Differential motor control enabled forward motion, reversing, and turning. This rear-wheel-drive layout represented a deliberate simplification from the all-wheel-drive concept originally selected in Milestone 1. The change was justified on the grounds that the operating surface was smooth and flat, making the additional traction and mechanical complexity of all-wheel drive unnecessary. The simplified layout reduced wiring complexity, current draw, and fabrication effort while maintaining adequate mobility for the defined arena conditions.

The chassis underwent three distinct design iterations during prototyping. The first iteration used a broad board-style structure with recessed indentations intended to simplify integration of electronic components, side panels, and the scoop assembly. While functional for early testing, this layout introduced unnecessary structural material and therefore excessive mass. A second iteration retained the indentation-based packaging concept but reduced overall thickness and consolidated the component arrangement into a tighter footprint. Although this improved packaging density, further testing during milestone validation revealed that the platform could be simplified more substantially. The final chassis adopted a skeletal structure retaining only the essential load-bearing geometry. Electronics were arranged compactly at the rear alongside the drive motors, and custom 3D-printed retaining pins secured components without additional fastener hardware. This progressive mass reduction improved the power-to-weight ratio of the robot and reduced the actuation load on the scoop-lifting servos.

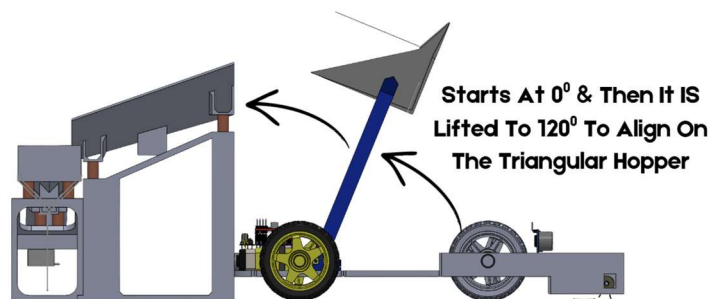


Figure 2: SolidWorks side-view assembly drawing of the scavenger robot showing the scoop mechanism in the raised (deposit) position. The arrow indicates the rotational travel of the servo-driven scoop arms from the lowered collection position to the raised unloading position.

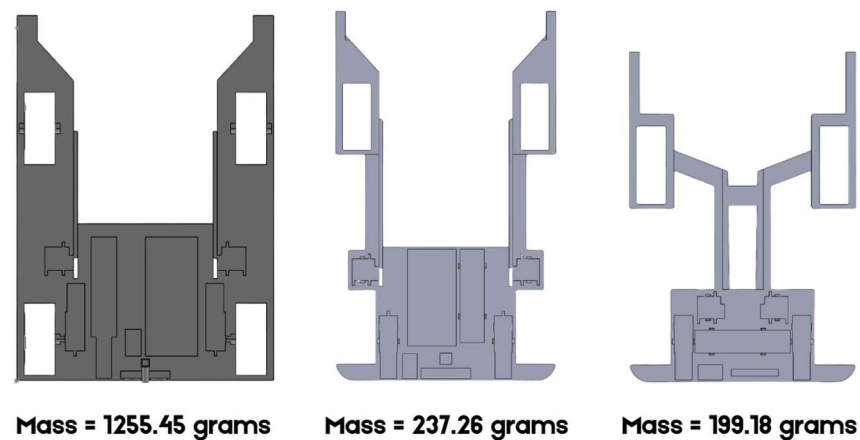


Figure 3: Chassis design evolution across three prototype iterations. Left: first-generation broad board chassis with recessed component indentations. Centre: second-generation reduced-thickness chassis with consolidated component layout. Right: final skeletal chassis retaining only essential load-bearing geometry, with 3D-printed retaining pins replacing conventional fastener hardware.

Mobile Robot — Object Collection

Object collection was achieved using a front scoop combined with an active rotary brush. The scoop provided a wide ground-level capture area and funneled erasers toward the intake region as the robot advanced. The brush, constructed from pipe-cleaner bristles and driven by a DC motor, rotated inward during forward travel to pull scattered erasers into the scoop more effectively than a passive plow alone. This arrangement increased collection probability and allowed multiple erasers to be gathered continuously during a single forward pass. During reverse motion, the scoop was raised and brush rotation was coordinated to retain collected objects and prevent them from escaping the scoop cavity. This coordinated scoop-and-brush interaction allowed the same front mechanism to serve both collection and containment functions, reducing the need for a separate internal transfer system.

The scoop mechanism also underwent substantial redesign between prototype iterations. Early versions used a wide rectangular scoop supported by arms mounted near the outer edges of the chassis. Testing revealed two significant problems with this arrangement. First, insufficient arm clearance caused the

scoop arms to rub against the chassis structure during lifting, imposing excessive frictional load on the servos and risking overheating or stall. Second, the scoop's own mass was high enough to challenge servo lifting capacity before any eraser payload was added. To address both issues, the scoop was redesigned to be significantly lighter and tapered to better match the triangular entry geometry of the sorting station, improving unloading efficiency and reducing the probability of erasers missing the vibratory feeder during the deposition step. The scoop arms were relocated inward toward the chassis centreline to eliminate wall contact during travel, and their geometry was adjusted to reduce the mechanical tension generated by an overly wide mounting span. A transverse connecting beam was added between the two arms, enabling both servo-driven sides to share the lifting load, improving symmetry of motion, and reducing torsional twisting under payload. Collectively, these changes reduced the effective torque demand on the servo system and improved the reliability of the lifting and unloading cycle.

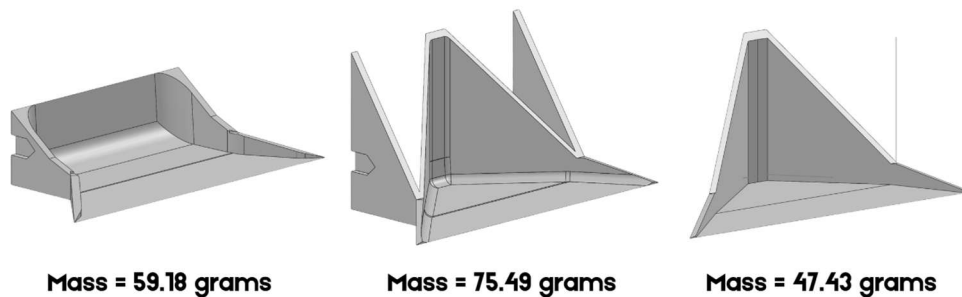


Figure 4: Scoop mechanism design evolution. Left: original wide rectangular scoop with outboard arm mounting, which caused chassis wall contact and excessive servo loading during lifting. Right: final tapered scoop profile with inboard arm relocation, transverse connecting beam, and reduced structural mass, restoring reliable servo operation within the rated torque margin.

Mobile Robot — Navigation and Return-to-Base

The robot performed search coverage under IMU-assisted heading control, with the strategy evolving across prototype stages. The initial lawnmower-style zig-zag pattern deposited too many erasers

simultaneously and overloaded the scoop servos, and was refined into the three-cycle fan-shaped path described below. In both implementations, the IMU was calibrated at the start of each run and heading corrections applied throughout motion to maintain directional consistency. A rear-mounted ultrasonic sensor detected proximity to the sorting station and triggered the stop-and-deposit sequence. A V-shaped alignment wall — two flat walls angled at 50 degrees — provided passive lateral correction during reverse approach: the filleted chassis corners contacted the walls and self-centred the robot without any active software correction.

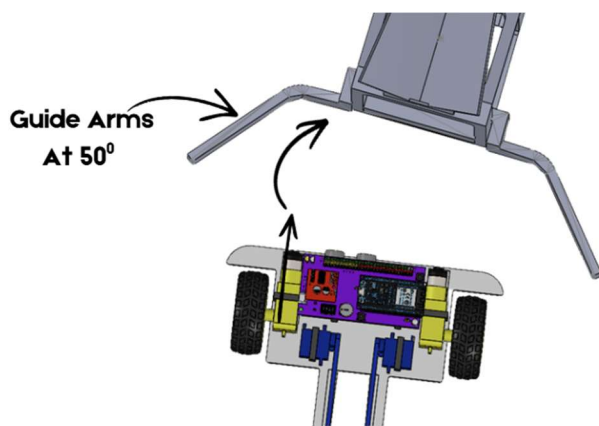


Figure 5: SolidWorks drawing of the V-shaped base station alignment walls, showing the two flat walls angled at 50 degrees forming a passive docking guide. The filleted chassis corners of the robot contact and slide along the walls during reverse approach, self-correcting any lateral offset until the robot is centred and the ultrasonic sensor confirms contact with the base.

Collection Strategy — Multi-Cycle Operation

The overall collection strategy was refined during integration testing from a single extended pass to a multi-cycle sequence of shorter collection-and-deposition runs. In the final control sequence, the robot began at the base station and executed three distinct collection cycles. Cycle 1 drove straight forward along the central corridor and reversed directly back to the base to deposit. Cycle 2 turned approximately 45 degrees to the right before advancing forward, then reversed back to the deposit point and realigned. Cycle 3 mirrored this on the left side. Together, the three cycles formed a fan-shaped coverage pattern across the full width of the arena, as shown in Figure 6. This staged approach offered two key engineering advantages. First, it allowed the sorting mechanism to begin processing earlier deposits while the robot continued collecting, improving the effective use of the 120-second mission window. Second, it capped the maximum scoop payload per cycle, reducing peak servo loading and preventing the sorter from being

overwhelmed by a large simultaneous deposit. In practice, this multi-cycle strategy improved both mechanical reliability and sorting throughput consistency.

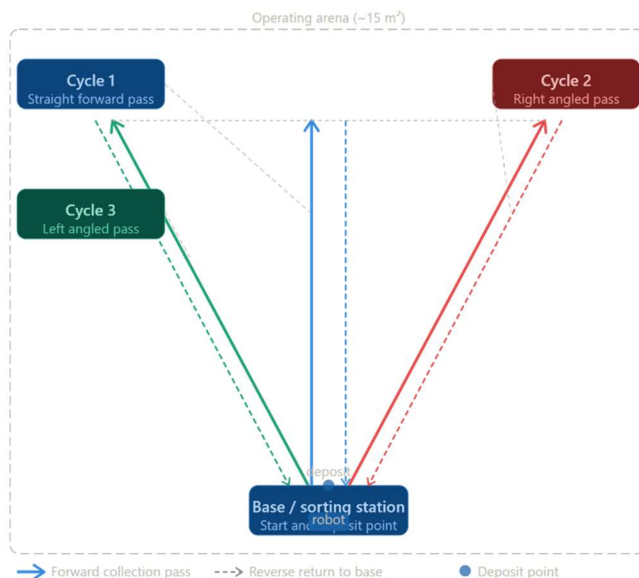


Figure 6: Motion sequence diagram showing the three-cycle collection path executed by the robot within the operating arena. Cycle 1 follows a straightforward-and-return path along the central corridor; Cycles 2 and 3 follow right and left diagonal paths respectively, forming a fan pattern that covers the full arena width. Solid lines indicate forward collection passes; dashed lines indicate reverse return to the base station.

Sorting System:

Once the mobile robot deposited its collected load at the base station, object classification and final deposition were handled entirely by the stationary sorting subsystem. The sorting station underwent a full architectural redesign between prototype stages after early testing revealed that the initial concept could not provide reliable single-object presentation to the sensor. The final design process therefore focused on selecting a sorting mechanism that could be fabricated with available resources, tuned experimentally, and integrated consistently with the unloading behaviour of the robot.

The initial sorting concept used a motorized rotary indexing disc mounted on an inclined tripod frame, with extruded radial ridges sized to capture one eraser at a time. As the disc rotated, erasers were intended to be captured in the inter-ridge pockets and indexed individually into a blast-gate assembly, where a TCS34725 colour sensor would trigger a servo-actuated chute to sort them left (non-target) or right (green). However, physical testing showed that this concept could not reliably isolate individual erasers. Due to the compliant surface and irregular orientation of the erasers, the ridges would sometimes capture

multiple objects or none. Without a controlled upstream feeding mechanism, the system could not guarantee one-at-a-time delivery and was therefore rejected in favour of a different architecture.



Figure 7: Initial rotary indexing disc sorting concept.

The final sorting station used a dual-stage vibratory feeder, a colour sensing stage, and a servo-actuated bifurcated output chute. Each of the two inclined funnel platforms was driven by a DC motor mounted beneath it and supported by stiff compression springs, which provided the restoring force for consistent oscillatory motion. The stock 400:1 gearboxes were removed from both motors so the shafts could rotate at higher speed, and an eccentric offset weight was attached to each shaft to generate high-frequency vibration. This vibration disaggregated clustered erasers and promoted progressive single-file advancement toward the sensing chute. The final sorting station was built primarily from 3D-printed structural components and integrated a TCS34725 colour sensor for object classification, two modified DC motors with eccentric masses for vibratory feeding, stiff compression springs for controlled oscillation, and a servo-actuated chute for final object diversion. These components were selected because they provided a practical balance between simplicity, tunability, and reliable operation within the available fabrication methods and project budget.

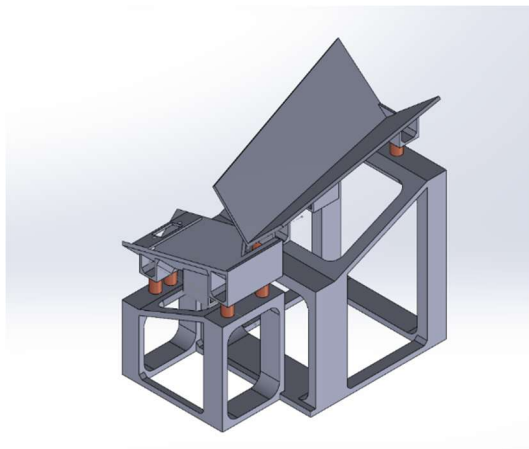


Figure 8: Final dual vibratory feeder sorting system.

Funnel angle proved to be a critical tuning parameter. Initial steep angles generated excessive eraser momentum and caused clustering at the sensing stage. After iterative testing, an angle of 16° from horizontal was established as optimal for both funnels, allowing gravity-driven advancement at a controlled rate. The two motors were operated in an alternating offset pattern, with the upper funnel active while the lower was inactive and

vice versa, so that erasers arrived at each stage with dissipated momentum. Both motors were driven at maximum PWM (255/255), which produced the most controlled and repeatable feeding behaviour.

At the base of the lower funnel, the TCS34725 colour sensor classified each eraser using calibrated RGB threshold values established during sensor testing. In the final implementation, an eraser was classified as green when the normalized green ratio exceeded 0.40 and the green-dominance term was positive. This rule was selected because the target green eraser produced a normalized green value of 0.4299 with positive dominance, while all non-target test objects produced lower normalized green values and non-positive dominance under the same sensing conditions. Objects that did not satisfy this condition were rejected by default, ensuring that sensor uncertainty resulted in missed green erasers rather than false acceptance of non-target objects.

Across 100 successive sorting trials, the final sorting system achieved 90–92% classification accuracy. The primary failure mode was eraser clustering at the sensing stage, where two objects entering simultaneously could produce a composite reading and occasional misclassification. A secondary limitation was base-station drift under sustained vibration, which was temporarily managed during the showcase by securing the unit to the surface. Despite these limitations, the achieved accuracy was considered acceptable for the scope of the project.

2.3 Design Analysis

The final design was evaluated from the perspectives of collection efficiency, mechanical feasibility, sorting reliability, and overall system integration. Given the academic project context, the analysis focused on whether the selected architecture achieved a realistic and defensible balance between performance and reliability, combining qualitative engineering reasoning with targeted quantitative assessments where practical measurements were available.

System-Level Architecture

The most consequential design decision was the separation of the full mission into two coordinated subsystems. By assigning collection, navigation, and transport exclusively to the mobile robot, and

classification and deposition exclusively to the stationary sorter, the architecture reduced the number of simultaneous sensing, actuation, and timing demands placed on any single unit. Relative to the onboard sorting architecture of Concept B, this arrangement was expected to yield a higher probability of mission completion within 120 seconds because the robot could optimise for collection speed without also managing real-time object discrimination. Relative to the passive gravity-fed sorting of Concept A, it offered greater control over object presentation to the colour sensor, since vibration-assisted feeding and active servo diversion are less sensitive to ramp angle tolerance and precise gravitational timing than a purely passive gate system. The sorting station itself also underwent a full internal architectural redesign between prototype stages, moving from a rotary indexing disc concept to a dual vibratory feeder arrangement, further demonstrating that the two-subsystem architecture provided the modularity to completely replace the sorting mechanism without any modification to the mobile robot — validating a key structural advantage of the selected concept. These comparisons support the conclusion that the selected architecture represented the most robust, modular, and controllable option among those evaluated.

Table 3: Quantitative and qualitative performance comparison between the three evaluated system-level concepts across key engineering criteria. Concept C is the only concept for which validated sorting accuracy data is available, as Concepts A and B were eliminated prior to prototype construction.

Criterion	Concept A	Concept B	Concept C (Selected)
Weighted decision score	3.15	2.75	4.2
Sorting method	Passive gravity gate	Onboard real-time sorting	Dual vibratory feeder
Navigation method	IR beacon homing	Encoder-based return	IMU-assisted heading
Collection method	Active scoop intake	Claw-based pickup	Scoop + brush intake
Sorting architecture iterations	1	1	2 (rotary disc → vibratory feeder)
Onboard mechanical complexity	Medium	High	Low–Medium
Jam risk assessment	High	Very high	Low
Sorting accuracy (validated)	Not tested	Not tested	90–92%
Fabrication feasibility	Medium	Low	High
Passed Go/No-Go screening	No	No	Yes

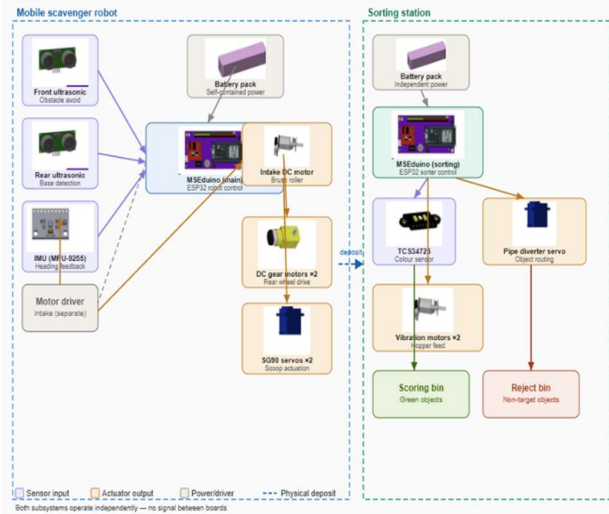


Figure 9: System block diagram of the autonomous scavenger system, showing sensor inputs, the ESP32-based controller, motor driver, actuator outputs, onboard power supply, and the object transfer path between the mobile robot and sorting station.

Mechanical Feasibility — Actuator Loading

A key mechanical concern throughout prototyping was the load imposed on the scoop-lifting servo system. The SG90 micro servo used for scoop actuation has a rated stall torque of 1.8 kg·cm (176.5 N·mm). Using the static torque relationship $T = mgL$, where m is the combined scoop and payload mass and L is the arm length from pivot to load point (178.5 mm), the torque demand of the original design was calculated as:

$$T_1 = (0.07549 + 0.018) \times 9.81 \times 0.1785 = \mathbf{0.1638 \text{ N}\cdot\text{m} (163.8 \text{ N}\cdot\text{mm})}$$

This represented 92.8% of the servo's stall torque — a critically small operating margin that directly explained the observed servo stall failures during early prototype testing. Following the scoop redesign, the reduced scoop mass of 47.43 g yielded a revised torque demand of:

$$T_2 = (0.04743 + 0.018) \times 9.81 \times 0.1785 = \mathbf{0.1146 \text{ N}\cdot\text{m} (114.6 \text{ N}\cdot\text{mm})}$$

This corresponds to 64.9% of stall torque, restoring a safe operating margin and representing a **30.0% reduction in torque demand** relative to the original design. The addition of the bilateral connecting beam between the scoop arms further distributed this load across both servos, improving symmetry and reducing the risk of asymmetric stall. These calculations confirm that the scoop redesign was not merely a

qualitative improvement but a quantitatively necessary correction to bring the actuation system within its reliable operating range.

Figure 10: Handwritten calculation of servo torque demand before and after scoop redesign. The original design operated at 92.8% of the SG90 stall torque rating, explaining observed servo stall failures. The redesigned scoop reduced torque demand to 64.9% of stall torque, representing a 30.0% reduction and restoring a reliable operating margin.

Sorting Station — Flow Control and Classification Confidence

The sorting station was analysed in terms of object flow regularity and classification reliability. As described in Section 2.2, the initial rotary disc was abandoned when eraser compliance prevented consistent single-object indexing. The dual vibratory feeder resolved this by converting bulk deposits into

1. Servo Torque Demand — Scoop Lifting
Given:
m_{s1} (original scoop) = 75.49 g = 0.07549 kg
m_{s2} (redesigned scoop) = 47.43 g = 0.04743 kg
m_{e3} (eraser payload, avg 9 erasers $\times$ 2 g) = 18 g = 0.018 kg
L (arm length, pivot to scoop load) = 178.5 mm = 0.1785 m
$g = 9.81 \text{ m/s}^2$
SG90 rated stall torque = 1.8 kg-cm = 176.5 N-mm
Torque formula: $T = (m_s + m_e) \times g \times L$
Original design:
$T_1 = (0.07549 + 0.018) \times 9.81 \times 0.1785$
$T_1 = 0.09349 \times 9.81 \times 0.1785$
$T_1 = 0.1638 \text{ N-m} = 163.8 \text{ N-mm}$
% of stall torque = $(163.8 / 176.5) \times 100 = 92.8\%$ Δ near stall limit
Redesigned scoop:
$T_2 = (0.04743 + 0.018) \times 9.81 \times 0.1785$
$T_2 = 0.06543 \times 9.81 \times 0.1785$
$T_2 = 0.1146 \text{ N-m} = 114.6 \text{ N-mm}$
% of stall torque = $(114.6 / 176.5) \times 100 = 64.9\%$ $\checkmark$ safe margin
Load reduction:
$\Delta T = (163.8 - 114.6) / 163.8 \times 100$
$\Delta T = 30.0\%$ reduction in torque

a controlled single-file stream before the TCS34725 sensing region. Classification confidence depended on calibrated funnel angle (16 degrees), alternating motor offset pattern, and maximum PWM (255/255). The fail-safe default of left-diversion for all unclassified objects ensured sensor failures produced missed green erasers rather than non-target contamination. Across 100 successive trials the system achieved 90–92% accuracy, with simultaneous dual-object arrival as the primary

failure mode.

Multi-Cycle Motion Strategy

The transition from the initial single-pass coverage strategy to the final multi-cycle fan-shaped path improved system performance in a measurable and quantifiable way. A cycle time budget was constructed from the measured durations of each operational phase, as shown in Figure 11. Cycle 1 (straight forward

pass) totalled 14,000 ms. Cycles 2 and 3 (right and left angled passes respectively) totalled 15,400 ms each, accounting for the additional in-place turn and realignment manoeuvre. The combined estimated robot motion time was therefore 44,800 ms (44.8 s). Across three timed full-system runs, the measured average completion time was $(87 + 87 + 83) / 3 = \mathbf{85.7 \text{ seconds}}$, comfortably within the 120-second mission limit and providing a working margin of 34.3 seconds. The difference between the estimated 44.8s and the measured 85.7s accounts for residual eraser sorting time after robot motion was complete, confirming that the throughput capacity of the sorting station — rather than robot travel time — was the primary determinant of overall cycle duration. This has an important design implication: further improvements to overall system performance would be more effectively achieved by increasing sorting throughput than by reducing robot travel time.

2. Mission Cycle Time Budget

Cycle 1 — straight forward collection pass:

$$t = 4000 + 1000 \text{ (scoop up)} + 4000 + 1000 \text{ (deposit)} + 4000 \text{ (sort wait)} = 14\,000 \text{ ms}$$

Cycle 2 — right-side curved pass:

$$t = 1200 \text{ (turn R)} + 4000 + 4000 \text{ (return)} + 1200 \text{ (realign)} + 1000 + 4000 = 15\,400 \text{ ms}$$

Cycle 3 — left-side curved pass (mirrored):

$$t = 1200 \text{ (turn L)} + 4000 + 4000 \text{ (return)} + 1200 \text{ (realign)} + 1000 + 4000 = 15\,400 \text{ ms}$$

Total robot motion + sorting:

$$t_{\text{total}} = 14\,000 + 15\,400 + 15\,400 = 44\,800 \text{ ms} \approx 44.8 \text{ s (estimated)}$$

Measured average across 3 timed runs:

$$\bar{t} = (87 + 87 + 83) / 3 = 257 / 3$$

$$\bar{t} = \mathbf{85.7 \text{ s} < 120 \text{ s limit} \checkmark (34.3 \text{ s margin})}$$

Estimated vs. measured difference = $85.7 - 44.8 = 40.9 \text{ s}$ accounts for remaining eraser sorting time after robot motion completes.

Figure 11: Handwritten cycle time budget

for the three-cycle fan-shaped collection sequence. Estimated motion time was 44.8 s, and the measured average completion time over 3 runs was 85.7s, leaving a 34.3s margin within the 120 s limit. The 40.9 s difference was due to sorting after motion ended.

Overall Confidence Assessment

Confidence in the final design is assessed as moderate to high. The strongest areas are: the two-subsystem architecture's demonstrated ability to accommodate a complete sorting mechanism replacement without modifying the robot; the quantitatively confirmed servo torque reduction from 92.8% to 64.9% of stall; and empirically validated 90–92% sorting accuracy. Confidence is lower in cycle-to-cycle repeatability because the system is sensitive to IMU zero-point drift, ultrasonic threshold, vibration intensity, and colour-sensor thresholds — all of which shift with battery state, lighting, or reassembly. These would be addressed in a future iteration through tighter tolerances and closed-loop feedback. Nevertheless, the

iterative refinement process — spanning three chassis versions, two scoop designs, two sorting architectures, and a navigation revision — provides clear evidence of rigorous engineering development.

2.4 Product Validation

Prototype validation used repeated subsystem and integrated testing to assess whether the design met its objectives. A total of 86 runs were completed during development, with the final 12 runs using fixed code and hardware reported here as the most representative of final performance.

Drive System and Navigation Validation

Drive system testing confirmed reliable forward motion, reversing, and turning under battery power with all mechanisms installed. IMU heading control produced acceptable directional consistency and the robot returned successfully to the base station across repeated trials. The V-shaped alignment walls compensated for small heading errors during approach. The rear-wheel-drive layout and IMU navigation strategy were confirmed sufficient for the arena conditions.

Collection Subsystem Validation

Collection testing confirmed that the original scoop geometry caused servo stall under load. Following the chassis and scoop redesigns, reducing torque demand from 92.8% to 64.9% of stall, lifting performance improved substantially and the robot deposited erasers consistently across trials. The tapered scoop also improved transfer alignment with the sorting station entry, reducing misaligned deposits.

Sorting Station Validation

Sorting station validation confirmed that excessive vibration amplitude or poor track geometry caused multiple erasers to arrive simultaneously. Iterative adjustment of motor speed and track angle improved single-object presentation. TCS34725 threshold values were established through calibration trials with both eraser types under representative lighting conditions, with the RGB ratio threshold set to maximise discrimination margin. These thresholds were applied consistently across all validation runs.

Integrated System Validation

Integrated testing confirmed that the robot and sorting station executed the complete operational sequence — collection, return, unloading, and classification — as one coordinated process. Across the 12 final runs, the system collected an average of 15–17 erasers per sequence, returned from Cycle 1 in all 12 runs, Cycle 2 in 10 of 12, and Cycle 3 in 8 of 12. The sorting station correctly classified green erasers in 11 of 12 runs; the single misclassification occurred when a green eraser immediately followed a non-target, producing a mixed reading. Isolated failures were caused by erasers lodging at the hopper entry or minor floor-drift. The system completed the task within 120 s in all 12 runs at an average of 85.7 s.

Table 4: Robot subsystem test plan and results summary.

Test objective	Method	Pass criterion	Result	Design impact
Drive forward and reverse reliably	Timed straight runs under full load	Consistent travel with no stall	Pass — consistent across all runs	Confirmed rear-wheel-drive layout sufficient
IMU heading consistency	Repeated forward passes measuring lateral drift	Drift < 15 cm over 1 m	Pass — acceptable drift for arena size	IMU calibration procedure standardised
Ultrasonic stopping accuracy	Reverse approach to station, measure stop distance	Stop within ± 3 cm of target	Pass — repeatable with fixed threshold	Threshold value fixed in final code
Scoop lifting under load	Lift loaded scoop (9 erasers) 5 consecutive times	No servo stall in 5/5 attempts	Pass after redesign; failed with original scoop	Triggered scoop mass reduction and arm geometry revision
Eraser capture rate	Forward pass over 10 distributed erasers	Capture $\geq 7/10$ erasers per pass	Pass — avg 8–9 per straight pass	Brush rotation speed confirmed effective
Eraser retention during reverse	Reverse 4,000 ms with loaded scoop raised	No erasers lost during transit	Pass — coordinated brush stop effective	Brush stop timing confirmed in code

Table 5: Sorting station test plan and results summary.

Test objective	Method	Pass criterion	Result	Design impact
Single-object presentation	Deposit 10 erasers, observe sensor region	$\geq 8/10$ arrive individually	Pass after track angle adjustment	Track angle and vibration intensity tuned
Green eraser classification	Present 20 green erasers individually	$\geq 19/20$ correctly identified	Pass — 20/20 in calibration trials	RGB threshold value fixed
Non-target rejection	Present 20 non-green erasers individually	$\geq 19/20$ correctly rejected	Pass — 20/20 in calibration trials	Threshold margin confirmed adequate
Mixed-colour stream sorting	Deposit mixed batch of 10 erasers	$\geq 8/10$ correctly sorted	Pass — 9/10 in mixed stream trials	One misclassification from close succession
Vibration feed continuity	Run station for 120 s continuously	No jams or feed stoppages	Pass — no mechanical stoppages observed	Vibration motor speed confirmed stable

Table 6: Integrated full-system validation results across 12 final runs using the fixed hardware and code configuration. The system completed the task within the 120-second limit in all runs, with an average completion time of 85.7 s.

Run	Erasers collected	Green correctly sorted	Non-targets rejected	Cycle 1 return	Cycle 2 return	Cycle 3 return	Time (s)	Notes
1	16	Yes	Yes	Yes	Yes	Yes	84	Clean run
2	17	Yes	Yes	Yes	Yes	Yes	83	Clean run
3	15	Yes	Yes	Yes	Yes	No	88	Cycle 3 drift
4	16	Yes	Yes	Yes	No	No	91	Eraser lodged at hopper
5	18	Yes	Yes	Yes	Yes	Yes	82	Best run
6	14	Partial	Yes	Yes	Yes	Yes	87	Green followed blue — mixed read
7	17	Yes	Yes	Yes	Yes	Yes	84	Clean run
8	15	Yes	Yes	Yes	Yes	No	86	Minor floor drift in Cycle 3
9	16	Yes	Yes	Yes	No	No	90	Two erasers lodged together at entry
10	17	Yes	Yes	Yes	Yes	Yes	83	Clean run
11	15	Yes	Yes	Yes	Yes	Yes	85	Clean run
12	16	Yes	Yes	Yes	Yes	Yes	84	Final validation run
Average / Summary	16	11/12 fully correct	12/12	12/12	10/12	8/12	85.7	Final fixed configuration



Figure 12: Photograph of the fully assembled autonomous scavenger system during final validation testing, showing the mobile robot positioned at the sorting station with the scoop in the raised deposit position and collected erasers entering the vibratory hopper tracks of the stationary sorting station.

Validation Limitations

The primary limitations of the validation process were prototype variability and the practical constraints of the course testing environment. The system remained sensitive to calibration-dependent parameters — specifically IMU zero-point drift, ultrasonic stopping threshold, vibration intensity, and colour-sensor thresholds — all of which could shift between sessions due to battery state, ambient lighting, or minor mechanical reassembly differences. Performance also varied with the initial spatial distribution of erasers within the arena, and the return success rate for Cycles 2 and 3 was lower than for Cycle 1, reflecting the greater accumulated heading error over longer and angled paths. As a result, the validation results should be interpreted as strong evidence of functional feasibility and meaningful improvement relative to earlier prototype iterations, rather than as demonstration of production-level repeatability. Even so, the testing programme provided clear and relevant evidence that the final design satisfied the primary project objectives and that the implemented refinements produced consistent and measurable improvements in collection, unloading, and sorting performance across all 12 final runs.

3. Conclusion

The final outcome of this project was a fully functional autonomous scavenger system composed of a mobile collection robot and a stationary sorting subsystem at the base station. The completed design satisfied the primary product specifications established at the start of the project: it operated

autonomously using an ESP32-based control system, collected multiple erasers from an unmarked floor area, returned to the base station without external guidance, sorted target objects from non-target objects, and completed the full operational sequence within the required 120-second time limit. Across the final 12 validation runs, the system completed the task in an average of 85.7 s, demonstrating that the selected architecture was feasible and effective within the given project constraints.

The two-subsystem architecture proved to be the most technically suitable solution among those evaluated. The ability to completely replace the sorting mechanism — from rotary disc to vibratory feeder — without modifying the mobile robot validated the modularity advantage of the selected concept. Key iterative improvements including the skeletal chassis, redesigned scoop, connecting beam, dual vibratory feeder, and multi-cycle fan path each produced measurable gains in reliability. These changes demonstrate that the final solution resulted from structured engineering refinement rather than trial and error.

The final design retained identifiable limitations: sensitivity to calibration-dependent parameters, lower return reliability for Cycles 2 and 3 due to accumulated heading error, and occasional sorting misclassification from simultaneous eraser arrival. These indicate the prototype remains a development-stage system with clear priorities for future refinement as described in Section 4.

4. Recommendations

Although the final prototype met the primary project objectives, testing identified several limitations that should be addressed in a future design iteration to improve reliability, repeatability, and overall scoring performance.

The first priority should be refinement of the collection mechanism. The scoop should be redesigned with taller side walls and a continuous funnel profile to retain objects more effectively, and the gap between the brush and scoop opening reduced, by repositioning the brush or adding a compliant rubber barrier, to

prevent re-ejection. These changes would improve object retention and increase erasers delivered per cycle.

The second priority should be improved return-path accuracy for Cycles 2 and 3. IMU-based control alone accumulated sufficient heading error over the angled paths to reduce docking reliability.

Supplementing IMU control with encoder feedback would allow simultaneous monitoring of angular drift and travel distance, reducing path deviation and improving base station docking consistency.

The most impactful sorting station improvement would be a single-file gating mechanism upstream of the colour sensor. Object clustering was the primary misclassification cause and the source of the 90–92% accuracy ceiling. A rotating escapement, timed flap, or oscillating gate — combined with a channel throat sized to the eraser geometry — would enforce one-eraser-at-a-time presentation and significantly reduce composite RGB readings.

The transfer geometry between the upper funnel and sensing chute should also be refined. Abrupt transitions promoted eraser overlap when large deposits were received. A smoother guided ramp with reduced drop height and better lateral constraint would dissipate momentum more gradually and improve spacing consistency before classification.

Finally, the base station should be made resistant to vibratory drift. A weighted base plate, wider footprint, or high-friction elastomeric pads would eliminate the unwanted station movement observed during operation and improve robot-to-station alignment consistency.

These five recommendations address the most significant remaining limitations: object loss during collection, return-path drift, multi-object sensing interference, abrupt sorter transitions, and base station instability. Implementing them in a future iteration would meaningfully improve collection reliability, sorting accuracy, and overall system robustness.

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Appendix A: Sensor Testing

Table D1: Sample Color Sensor Readings

Colour	R	G	B	C	rn	gn	bn	Hue (deg)	Dominance	Detected
Empty chute (black)	19	15	14	43	0.3958	0.3125	0.2917	12.0	-0.0833	No
Orange	164	55	61	256	0.5857	0.1964	0.2179	356.7	-0.3893	No
Blue	29	46	60	124	0.2148	0.3407	0.4444	207.1	-0.1037	No
Purple	45	31	39	104	0.3913	0.2696	0.3391	325.7	-0.1217	No
Lime	122	121	91	311	0.3653	0.3623	0.2725	58.1	-0.0030	No
Pink	154	58	76	263	0.5347	0.2014	0.2639	348.8	-0.3333	No
Green (target)	91	135	88	294	0.2898	0.4299	0.2803	116.2	0.1401	Yes

Based on the results, green erasers consistently produced higher normalized green values compared to other objects. Thresholds were selected to distinguish green objects from non-target objects using this difference. Testing confirmed that the selected thresholds allowed for consistent classification of objects under controlled conditions.

Appendix B: Bill of Materials (BOM)

	#	Item Description	Units	Quantity	Unit cost	Cost	Supplier	Part Number
Electronics	1	DC Electric Motor With Gear Box	EA	2	3.77	\$7.54	Temu	601101171782206
	2	MSEduino Board	EA	2	0.00	\$0.00	Custom PCB	
	3	Micro Servo Motors	EA	4	2.87	\$11.48	Aliexpress	1005005156530155
	4	Power Bank Battery	EA	2	11.68	\$23.36	Aliexpress	1005011764550161
	5	Breadboard	EA	2	1.79	\$3.57	Temu	601099605660774
	6	IMU	EA	1	3.71	\$3.71	Temu	601104528906315
	7	DC Motors	EA	1	4.77	\$4.77	Aliexpress	1005007009329460
Sensors	8	TCS3472 Color Sensor	EA	1	1.59	\$1.59	Aliexpress	1005007560583027
	9	Ultrasonic Sensor	EA	2	0.80	\$1.59	Aliexpress	1005007865485289
Mechanical Components	10	Roller Brush	EA	1	13.29	\$13.29	Amazon	B0D1RJ44JF
	11	A4988 Stepper Motor Driver	EA	1	1.59	\$1.59	Aliexpress	1005005980170404
	12	Mini Stepping Motor With Gearbox	EA	1	1.59	\$1.59	Aliexpress	1005009296243347
3D Printed Parts	13	Funnel	EA	1	0.00	\$0.00	Team Manufactured	
	14	Base	EA	1	0.00	\$0.00	Team Manufactured	
	15	Scoop	EA	1	0.00	\$0.00	Team Manufactured	
	16	Bucket Arm	EA	2	0.00	\$0.00	Team Manufactured	
	17	Sorting Tube	EA	3	0.00	\$0.00	Team Manufactured	
	18	Trap Door	EA	1	0.00	\$0.00	Team Manufactured	
					TOTAL	\$74.08		

The bill of materials (BOM) summarizes the components used in the system along with their quantities and estimated costs.